DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	S-ITNT416
COURSE TITLE	**:**	Big Data And Analytics
COURSE TYPE	**:**	Lecture
COURSE CREDIT	**:**	2 Units
LEARNING MODALITY	**:**	Traditional Blended Learning Model
PRE-REQUISITES	**:**	S-ITNT344LC
CO-REQUISITES	**:**	S-ITNT416LA
PROFESSOR	**:**	Paulino H. Gatpandan, DIT phgatpandan@dlsud.edu.ph
CONSULTATION HOURS	**:**	1000 – 1200 Tuesday (COS100C) 1000 – 1200 Wednesday (COS100C) 0900 – 1200 & 1300 – 1430 Friday (Online)

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Provides students with an overview of the current trends in information technology that drives today's business. The course provides understanding on data management techniques that can help an organization achieve its business goals and address operational challenges. This also introduces different tools and methods used in business analytics to provide the students with opportunities to apply these techniques in simulations in a computer laboratory.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO1. Explain the characteristics of Big Data.
CLO2. Explain the role and importance of each type of analytics in deriving insights from Big Data.

CLO3. Create data visualizations and dashboards to communicate analytical results effectively.
CLO4. Analyze datasets using machine learning algorithms and tools to extract meaningful insights.
CLO5. Explain the purpose and activities involved in each stage of the life cycle.
CLO6. Assess the impact of data governance and ethical considerations on Big Data projects.
CLO7. Explain how the trends are shaping the future of Big Data and analytics.
CLO8. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 1. Explain the characteristics of Big Data.	TLO1. Define Big Data and its characteristics. TLO2. Discuss how the characteristics impact the processing of Big Data.
CLO 2. Explain the role and importance of each type of analytics in deriving insights from Big Data.	TLO3. Define and differentiate descriptive, diagnostic, predictive, and prescriptive analytics. TLO4. Understand the role of each type of analytics.
CLO 3. Create data visualizations and dashboards to communicate analytical results effectively.	TLO5. Demonstrate proficiency in using data visualization tools. TLO6. Design clear and informative visualizations that accurately represent data insights.
CLO 4. Analyze datasets using machine learning algorithms and tools to extract meaningful insights.	TLO7. Preprocess datasets by cleaning, normalizing, and transforming data to be suitable for machine learning models. TLO8. Demonstrate the practical application of machine learning models to derive actionable insights from various datasets.
CLO 8. Demonstrate 21st century skills in all learning activities.	TLO13. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Introduction Module **Module 1** CLO1 CLO8 TLO1 TLO2 TLO13	Week 1 Aug 19-24	**Course Orientation** • Gospel Reading, Reflection and Prayer • Course Overview • Course Polices and Requirements • Professor Profile • Grading System **Introduction to Big Data, Data Generation and Acquisition** • Introduction to Big Data • Characteristics of Big Data • Data Sources and Types • Data Acquisition and Methods • Data Quality and Preprocessing • **Enabling Assessment:** Quiz on Module 1	Schoolbook	/		2.0
Module 1 CLO1 TLO1 TLO2	Week 2 Aug 26-31	**Introduction to Big Data, Data Generation and Acquisition** (continuation) • Gospel Reading, Reflection and Prayer	Power Point		/	2.0
Module 2 CLO2 TLO3 TLO4	Week 3 Sep 2-7	**Big Data Storage and Management, and Types of Analytics** • Gospel Reading, Reflection and Prayer • Traditional Databases vs Big Data Storage Solutions • No SQL databases • Distributed File Sytems • Types of Analytics	Power Point	/		2.0

Module 2 CLO2 CLO8 TLO3 TLO4 TLO13	Week 4 Sep 9-14	**Big Data Storage and Management, and Types of Analytics** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** Quiz on Module 2	Schoolbook		/	2.0
Module 3 CLO3 TLO5 TLO6	Week 5 Sep 16-21	**Data Warehousing and OLAP** • Gospel Reading, Reflection and Prayer • Data Warehousing Concepts • Online Analytical Processing • Tools for Data Warehousing and OLAP	Power Point	/		2.0
Module 3 CLO3 CLO8 TLO5 TLO6 TLO13	Week 6 Sep 23-28	**Data Warehousing and OLAP** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** Group Activity: Each group will design a data warehouse schema for their assigned business domain	Schoolbook		/	2.0
Module 4 CLO3 CLO4 CLO8 TLO5 TLO6 TLO7 TLO8 TLO13	Week 7 Sep 30 - Oct 5	**Data Analysis and Visualization, and Machine Learning with Big Data** • Gospel Reading, Reflection and Prayer • Exploratory Data Analysis • Data Visualization • Tools for Data Visualization • Introduction to Machine Learning • Machine Learning with Big Data • **Enabling Assessment:** Quiz on Module 4	Schoolbook		/	2.0
Module 4 CLO3 CLO4 TLO5 TLO6 TLO7 TLO8	Week 8 Oct 7-12	**Data Analysis and Visualization, and Machine Learning with Big Data** (continuation) • Gospel Reading, Reflection and Prayer	Power Point	/		2.0

	Week 9 Oct 14-19	**Summative Assessment** Midterm Examination		/		2.0
			Total	**10.0**	**8.0**	**18.0**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO 5. Explain the purpose and activities involved in each stage of the life cycle.	TLO9. Describe the stages of the Big Data analytics life cycle, including data collection, processing, analysis, and visualization.
CLO 6. Assess the impact of data governance and ethical considerations on Big Data projects.	TLO10. Define ethical issues related to Big Data, including data privacy, consent, transparency, and bias. TLO11. Analyze how data governance and ethical practices influence the success and reliability of Big Data projects.
CLO 7. Explain how the trends are shaping the future of Big Data and analytics.	TLO12. Identify the challenges and potential risks associated with adopting new trends in Big Data and analytics.
CLO 8. Demonstrate 21st century skills in all learning activities.	TLO13. Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)

Module Number and Learning Outcome/s	Date	Teaching-Learning Lessons and Activities / Assessment Strategies	Technology Enabler	Onsite	Asynchronous	Allotted Hours
Module 5 CLO5 CLO8 TLO9 TLO13	Week 10 Oct 21-26	**Big Data Analytics Life Cycle, and Big Data Platforms** • Gospel Reading, Reflection and Prayer • Big Data Analytics Life Cycle • Ethics and Privacy in Big Data Analytics • Big Data Platforms and Tools • **Enabling Assessment:** Quiz on Module 5	Power Point	/		2.0

	Week 11 Oct 28 – Nov 2	Independent Learning Week	Schoolbook		/	2.0
Module 6 CLO6 TLO10 TLO11	Week 12 Nov 4-9	**Big Data Security and Governance** • Gospel Reading, Reflection and Prayer • Big Data Security: Challenges and Solutions • Data Governance • Regulatory Compliance and Data Protection Laws	Power Point	/		2.0
Module 6 CLO6 CLO8 TLO10 TLO11 TLO13	Week 13 Nov 11-16	**Big Data Security and Governance** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** Quiz on Module 6	Schoolbook		/	2.0
Module 7 CLO6 CLO7 CLO8 TLO10 TLO11 TLO12 TLO13	Week 14 Nov 18-23	**Industry Applications of Big Data** • Gospel Reading, Reflection and Prayer • Big Data in Healthcare • Big Data in Finance • Big Data in Retail • **Enabling Assessment:** Case Study on the applications of Big Data	Schoolbook		/	2.0
Module 7 CLO6 CLO7 TLO10 TLO11 TLO12	Week 15 Nov 25-30	**Industry Applications of Big Data** (continuation) • Gospel Reading, Reflection and Prayer	Power Point	/		2.0
Module 8 CLO7 TLO12	Week 16 Dec 2-7	**Emerging Trends in Big Data** • Gospel Reading, Reflection and Prayer • Internet of Things (IoT) and Big Data • Artificial Intelligence and Big Data • Future Trends: Quantum Computing and Big Data	Power Point	/		2.0

Module 8 CLO7 CLO8 TLO12 TLO13	Week 17 Dec 9-14	**Emerging Trends in Big Data** (continuation) • Gospel Reading, Reflection and Prayer • **Enabling Assessment:** Journal review of published papers of faculty members indicated in the references	Schoolbook		/	2.0
	Week 18 Dec 16-18	**Summative Assessment** Final Examination		/		2.0
			Total	**10.0**	**8.0**	**18.0**

GRADING SYSTEM:

Midterm

Class Participation – 20%
Enabling Assessments – 50%
Summative Assessments – 30%
100%

Finals

Class Participation – 20%
Enabling Assessments – 50%
Summative Assessments – 30%
100%

Overall Final Grade = (Midterm + Finals) / 2

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum

response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:

i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
iii. Email – you may email your professor. (Professor's email will be given during the course orientation)

c. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones cannot also be used as calculator during examination.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES:

Call number or e-provider	References
HD 69.B87.M184 2023	Hemachandran K, et al. 2023. Machine learning for business analytics: real-time data analysis for decision-making. New York: Routledge
QA 76.9.B45 .C378 2021	Castellani B, Rajaram R. 2021. Big data mining and complexity. London: SAGE Publications
HD 30.27 .M299 2019	Malviya A, Malmgren M. 2019. Big data for managers: creating value. London; New York: Routledge, Taylor and Francis group
QA 76.9.D26 .C816 2019	Coronel C, Morris S. 2019. Database systems: design, implementation and management. 13th edition. (Australia, United States): Cengage Learning
Journals	Z. Gao, M. P. Gatpandan and P. H. Gatpandan, "Classification Decision Tree Algorithm in Predicting Students' Course Preference," *2021 2nd International Symposium on Computer Engineering and Intelligent Communications (ISCEIC)*, Nanjing, China, 2021, pp. 93-97, doi: 10.1109/ISCEIC53685.2021.00026. J. Xu, P. H. Gatpandan, M. P. Gatpandan and Z. Gao, "Freight and Passenger Volume Prediction using Multiple Regression," *2021 International Conference on Machine Learning and Intelligent Systems Engineering (MLISE)*, Chongqing, China, 2021, pp. 77-83, doi: 10.1109/MLISE54096.2021.00022. X. Gao, M. F. Rosas, Y. Zhang, M. P. Gatpandan, P. H. Gatpandan and D. Dasig, "The Research and Analysis of Student Health Status on the Basis of Physic Fitness Test' Data," *2021 International Conference on Public Health and Data Science (ICPHDS)*, Chengdu, China, 2021, pp. 110-115, doi: 10.1109/ICPHDS53608.2021.00031. D. J. R. Calantoc, D. D. Dasig, E. E. Claricia, M. A. B. Taduyo, P. H. Gatpandan and R. V. F. Guarin, "Clustering and Predicting Smartphones Features using Gaussian Mixture Model Algorithm," *2021 IEEE 13th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*, Manila, Philippines, 2021, pp. 1-6, doi: 10.1109/HNICEM54116.2021.9731964.

Prepared: **Information Technology Department**
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Approved: **(SGD) Ms. Sherry B. Naz**
Chair, Information Technology Department

Endorsed: **(SGD) Ms. Marivic R. Mitschek**
Dean, College of Information and Computer Studies

cf: OVCAR, AEA, CICS Curriculum Development Committee, file